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4.7 Organic Chemistry

AQA GCSE Chemistry (8462) — Paper 2

Organic chemistry is the chemistry of **carbon compounds**. Carbon atoms can form four covalent bonds and can join to each other to make chains and rings of almost any length, so there are millions of organic compounds — far more than the compounds of every other element put together. This topic covers crude oil and the fuels made from it, cracking, the homologous series of alkenes, alcohols and carboxylic acids, and both synthetic and natural polymers. Sections marked **(Chemistry only)** are not in Combined Science; content marked **(HT only)** is Higher Tier.

4.7.1 Carbon Compounds as Fuels and Feedstock

4.7.1.1 Crude Oil, Hydrocarbons and Alkanes

> **Crude oil** is a **finite resource** found in rocks. It is the remains of an ancient biomass consisting mainly of **plankton** that was buried in mud millions of years ago and subjected to heat and pressure.
>
> Crude oil is a **mixture** of a very large number of compounds. Most of the compounds in crude oil are **hydrocarbons**.

> A **hydrocarbon** is a compound made of **hydrogen and carbon atoms only**.

Because crude oil is a *mixture* (not a compound), the substances in it are not chemically bonded to one another. Each keeps its own properties — including its own boiling point — which is why the mixture can be separated by a physical process (fractional distillation).

Most of the hydrocarbons in crude oil are **alkanes**. The alkanes are a **homologous series**.

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A **homologous series** is a family of compounds that:

- have the same **general formula**;
- differ from the next member by a **CH_2** unit;
- have the same **functional group** and so similar **chemical** reactions;
- show a **gradual trend** in physical properties (e.g. boiling point rises as the chain gets longer).

Alkanes have the general formula C_nH_{2n+2}.

They are **saturated** hydrocarbons — every carbon atom is joined to four other atoms by **single covalent bonds**; there are no C=C double bonds. This makes alkanes fairly unreactive (apart from combustion).

The First Four Alkanes

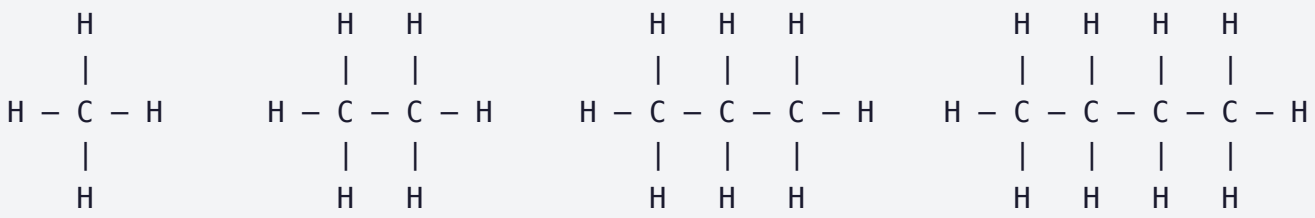
Name	Molecular formula	Condensed structural formula	Boiling point	State at room temperature
Methane	CH_4	CH_4	−162 °C	Gas
Ethane	C_2H_6	CH_3CH_3	−89 °C	Gas
Propane	C_3H_8	$CH_3CH_2CH_3$	−42 °C	Gas
Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$	−1 °C	Gas

A useful memory aid for the prefixes: **M**onkeys **E**at **P**eanut **B**utter — **meth-** (1 C), **eth-** (2 C), **prop-** (3 C), **but-** (4 C). The same prefixes are used for alkenes, alcohols and carboxylic acids, with **pent-** for 5 carbons.

Displayed Formulae of the First Four Alkanes

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A **displayed formula** shows every atom and every bond as a line. You must be able to recognise and draw these for methane, ethane, propane and butane.



> **Exam Tip:** Check a displayed formula by counting bonds: every **carbon must have exactly four bonds** and every **hydrogen exactly one**. Then check the formula fits C_nH_{2n+2}. For example, an alkane with 8 carbons has 2(8) + 2 = 18 hydrogens: C_8H_{18} (octane, a major component of petrol).

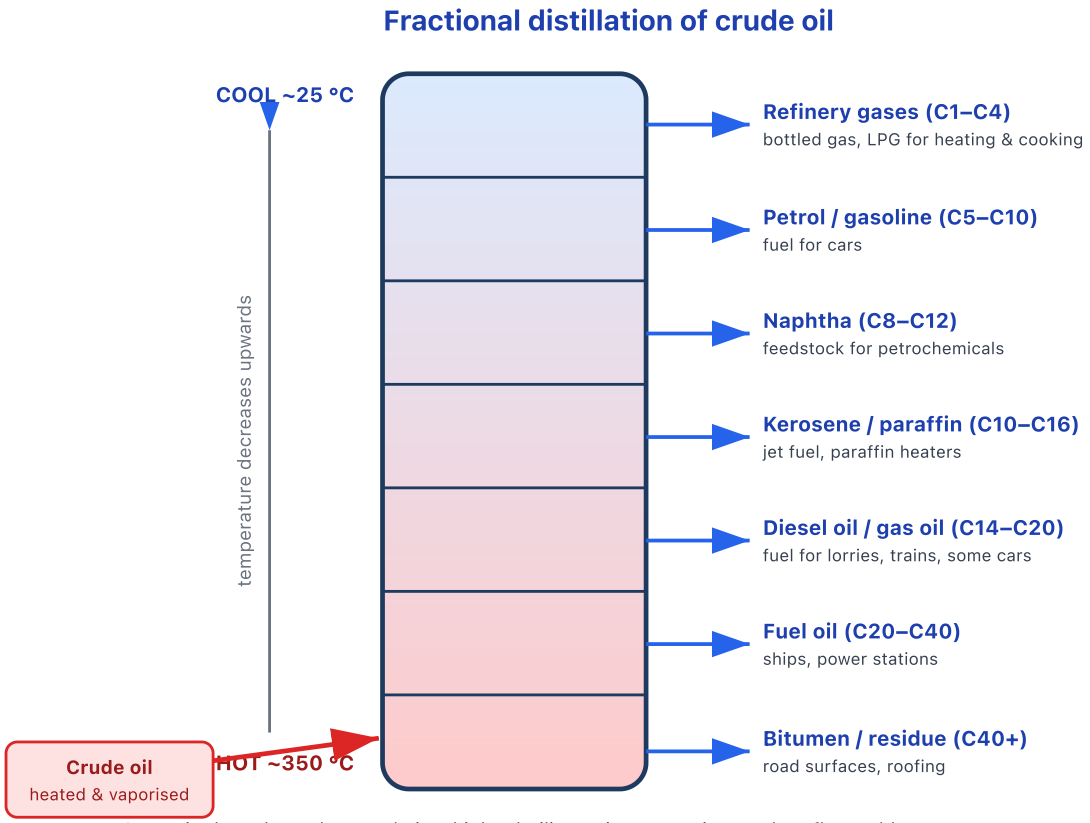
4.7.1.2 Fractional Distillation and Petrochemicals

The many hydrocarbons in crude oil can be separated into **fractions**, each of which contains molecules with a **similar number of carbon atoms** and therefore similar boiling points. This is done by **fractional distillation**.

How Fractional Distillation Works

1. Crude oil is **heated** to about 350–400 °C so that most of it **evaporates** (vaporises).
2. The vapours enter the bottom of a tall **fractionating column**, which has a **temperature gradient** — it is **hottest at the bottom** and **coolest at the top**.
3. The vapours **rise** up the column through trays fitted with bubble caps.
4. As the vapours rise, they cool. Each hydrocarbon **condenses** when it reaches a level where the temperature is **below its boiling point**; the liquid is collected from that tray.
5. **Long-chain** hydrocarbons have high boiling points, so they condense **low down** in the column (or never evaporate and are drained from the bottom as a residue).
6. **Short-chain** hydrocarbons have low boiling points, so they rise to the **top** and leave as gases.

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Lower in the column: longer chains, higher boiling point, more viscous, less flammable

Uses of Fractions — Fuels and Feedstock

Many of the fractions are used as **fuels**. The modern way of life depends on fuels from crude oil — petrol, diesel oil, kerosene, heavy fuel oil and liquefied petroleum gases.

Fractions are also processed in the **petrochemical industry** to produce **feedstock** (raw materials) for making many other useful materials, such as:

- **solvents** (e.g. for paints and glues);
- **lubricants** (engine oils, greases);
- **polymers** (plastics such as poly(ethene) and poly(propene));
- **detergents**.

The vast array of natural and synthetic carbon compounds exists because carbon atoms can form **families of similar compounds** — chains, branched chains and rings.

> **Exam Tip:** When asked to *explain* how fractional distillation works, you must mention all of: **heat/evaporate** the crude oil; **temperature gradient** (hot at the bottom, cool at the top);

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vapours **rise and cool**; each fraction **condenses at its boiling point** and is collected at a different height. "The oil is separated by boiling points" on its own scores very little.

4.7.1.3 Properties of Hydrocarbons

Some properties of hydrocarbons depend on the **size of their molecules** (the length of the carbon chain). These properties influence how hydrocarbons are used as fuels.

Property	Trend as chain length **increases**	Reason
Boiling point	Increases	Larger molecules have stronger **intermolecular forces** between them, so more energy is needed to separate the molecules.
Viscosity (how thick / slow-flowing a liquid is)	Increases	Long chains have stronger intermolecular forces and can tangle, so they flow less easily.
Flammability (how easily it ignites)	Decreases	Long-chain molecules are less volatile — fewer molecules evaporate to mix with oxygen and catch fire.
Volatility (how readily it evaporates)	Decreases	Higher boiling point means fewer molecules escape from the liquid at a given temperature.

So short-chain hydrocarbons make the best fuels — they are **runny, volatile and very flammable** with clean flames. Long-chain fractions are thick, hard to ignite and burn with a smoky flame, which is why there is a much greater demand for short-chain fractions than for long-chain ones.

Exam Tip: Do not say that boiling point increases because "the covalent bonds get stronger" — boiling a hydrocarbon does **not** break any covalent bonds. It overcomes the weak **intermolecular forces** between molecules, and these get stronger as the molecules get bigger. This is one of the most frequently penalised errors in the whole of GCSE Chemistry.

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Combustion of Hydrocarbon Fuels

The combustion of hydrocarbon fuels releases energy (it is **exothermic**). During combustion, the carbon and hydrogen in the fuel are **oxidised**.

> **Complete combustion** (plenty of oxygen) of a hydrocarbon produces **carbon dioxide and water**:
>
> `hydrocarbon + oxygen → carbon dioxide + water`

Examples you should be able to balance:

```
methane:   CH₄  + 2O₂   →  CO₂  + 2H₂O
propane:   C₃H₈ + 5O₂   →  3CO₂ + 4H₂O
butane:   2C₄H₁₀ + 13O₂ →  8CO₂ + 10H₂O
octane:   2C₈H₁₈ + 25O₂ → 16CO₂ + 18H₂O
```

Worked Example 1 — Balancing a Combustion Equation

```
Question: Write a balanced equation for the complete combustion
          of pentane, C₅H₁₂.

Step 1:  Carbon  — 5 C on the left, so 5CO₂ on the right.
Step 2:  Hydrogen — 12 H on the left, so 6H₂O on the right.
Step 3:  Oxygen on the right = (5 × 2) + (6 × 1) = 16 O atoms
         → 8 O₂ molecules needed on the left.

Answer:  C₅H₁₂ + 8O₂ → 5CO₂ + 6H₂O

Shortcut for alkanes CₙH₂ₙ₊₂: O₂ needed = n + (n+1)/2.
If this is a half number, double everything (as for butane above).
```

Incomplete combustion (when oxygen is limited) produces **carbon monoxide** and/or **soot** (carbon particles) as well as water — this is covered in 4.9 Chemistry of the Atmosphere.

4.7.1.4 Cracking and Alkenes

Fractional distillation produces far more long-chain hydrocarbons than are needed, and not enough of the short-chain fractions (such as petrol) that are in high demand. **Cracking** solves this problem.

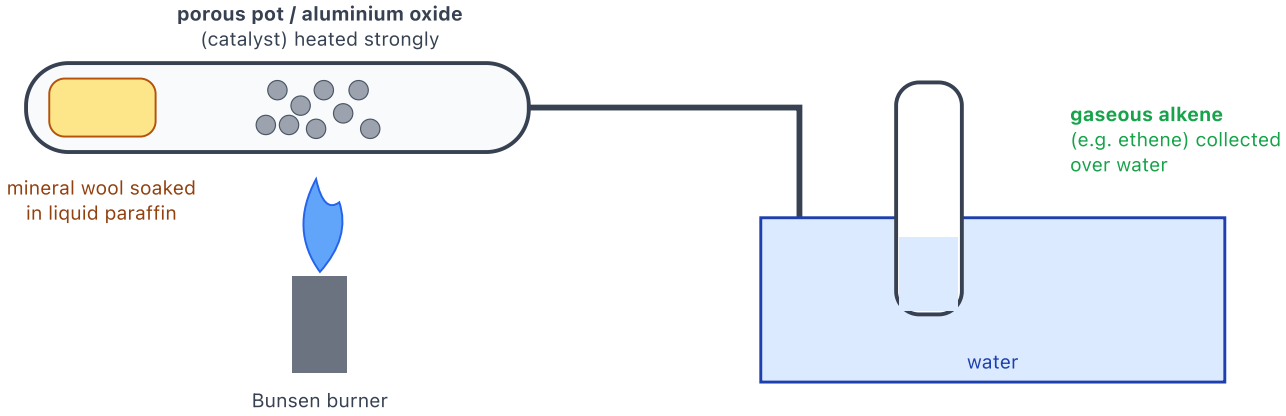
> **Cracking** is the breaking down of **long-chain hydrocarbons** into **shorter, more useful molecules**. It is a **thermal decomposition** reaction. The products of cracking include **alkanes** and another type of hydrocarbon called **alkenes**.

Two Methods of Cracking

Method	Conditions
Catalytic cracking	The heavy fraction is **vaporised** and the vapour is passed over a **hot catalyst** (a powdered zeolite or aluminium oxide) at around 550 °C.
Steam cracking	The vaporised hydrocarbons are **mixed with steam** and heated to a **very high temperature** (around 800–850 °C). No catalyst is needed.

Cracking liquid paraffin in the laboratory

Remove the delivery tube from the water before stopping heating, to prevent suck-back.



Writing Cracking Equations

In any cracking equation, the **number of carbon atoms and hydrogen atoms must balance**. Typically one product is a shorter alkane (C_nH_{2n+2}) and at least one is an alkene (C_nH_{2n}).

$C_{10}H_{22}$ → C_8H_{18} + C_2H_4
decane (alkane) — octane (alkane, for petrol) — ethene (alkene, used to make poly(ethene))

$C_{12}H_{26}$ → C_6H_{14} + C_4H_8 + C_2H_4
butene — ethene

Worked Example 2 — Completing a Cracking Equation

Question: Hexadecane, $C_{16}H_{34}$, is cracked to give octane, C_8H_{18}, and one other product. Identify the other product.

Carbon: 16 − 8 = 8
Hydrogen: 34 − 18 = 16
Other product = C_8H_{16}

C_8H_{16} fits C_nH_{2n} (16 = 2 × 8), so it is an ALKENE — octene.

Equation: $C_{16}H_{34} \rightarrow C_8H_{18} + C_8H_{16}$

Alkenes and the Test for Alkenes

Alkenes are more reactive than alkanes because they contain a carbon–carbon double bond. This allows them to be identified using **bromine water**.

Test for alkenes: shake the substance with **bromine water**.
- **Alkene** present: bromine water turns from **orange to colourless** (it is decolourised).
- **Alkane**: bromine water **stays orange** (no reaction).

Exam Tip: The colour change is **orange to colourless**. Never write "clear" — clear means transparent, and orange bromine water is already clear. Writing "clear" loses the mark every time.

Why Cracking Is Useful

- It converts low-demand long-chain fractions into **high-demand** shorter alkanes such as **petrol** — matching supply to demand.
- The **alkenes** produced are used to make **polymers** (e.g. ethene → poly(ethene); propene → poly(propene)) and as **starting materials** for the production of many other chemicals (e.g. ethanol by hydration of ethene).
- Students should be able to use data on supply and demand to explain why cracking is carried out.

Worked Example 3 — Supply and Demand

```
Fraction        Supply from crude oil (%)    Demand (%)
Gases                     2                      4
Petrol                   16                     27
Kerosene                 13                      8
Diesel                   19                     23
Fuel oil & bitumen       50                     38

Question: Use the data to explain why cracking is needed.

Answer: Demand for petrol (27%) is greater than the supply from
        fractional distillation (16%). Supply of fuel oil and
        bitumen (50%) exceeds demand (38%). Cracking converts the
        surplus long-chain fuel oil into shorter-chain molecules
        such as petrol, so supply better matches demand. It also
        produces alkenes needed to make polymers.
```

(Figures are illustrative, in the style of exam data questions.)

4.7.2 Reactions of Alkenes and Alcohols (Chemistry only)

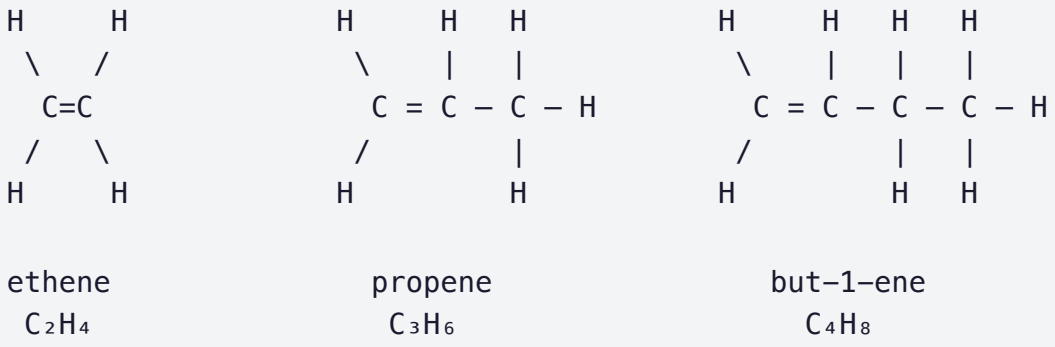
4.7.2.1 Structure and Formulae of Alkenes

> **Alkenes** are hydrocarbons with a **double carbon–carbon bond** (C=C). Their general formula is C_nH_{2n}.

Alkene molecules are **unsaturated** because they contain **two fewer hydrogen atoms** than the alkane with the same number of carbon atoms. The C=C double bond is the **functional group** of the alkenes.

Name	Molecular formula	Condensed structural formula
Ethene	C_2H_4	$CH_2{=}CH_2$
Propene	C_3H_6	$CH_2{=}CHCH_3$
Butene	C_4H_8	$CH_2{=}CHCH_2CH_3$ (but-1-ene) or $CH_3CH{=}CHCH_3$ (but-2-ene)
Pentene	C_5H_{10}	$CH_2{=}CHCH_2CH_2CH_3$ (pent-1-ene)

There is no "methene" — you need at least two carbon atoms to make a C=C double bond.



Exam Tip: For GCSE, the position of the double bond in butene and pentene is not normally required — "butene" is accepted. But your displayed formula must still have **four bonds on every carbon**: the two carbons in the C=C each have only two other bonds.

4.7.2.2 Reactions of Alkenes

Alkenes are **hydrocarbons**, so they burn in oxygen; and because they contain the reactive C=C double bond they undergo **addition reactions**.

Combustion of Alkenes

```
C₂H₄ + 3O₂ → 2CO₂ + 2H₂O        (complete combustion of ethene)
```

Alkenes tend to burn in air with **smoky (sooty) yellow flames** because they usually undergo **incomplete combustion** — they have a higher proportion of carbon than the corresponding alkanes, so there is often not enough oxygen to oxidise all the carbon.

Addition Reactions

> In an **addition reaction**, atoms are added across the C=C double bond. The double bond **opens up** to become a **single bond**, and one new atom (or group) joins to **each** of the two carbon atoms. Only **one product** is formed.

Reactant added	Conditions	Product	Example with ethene
Hydrogen (H_2) — *hydrogenation*	~60 °C, **nickel catalyst**	Alkane	$C_2H_4 + H_2 \rightarrow C_2H_6$ (ethane)
Water as steam (H_2O) — *hydration*	~300 °C, ~60–70 atm, **phosphoric acid catalyst**	Alcohol	$C_2H_4 + H_2O \rightarrow C_2H_5OH$ (ethanol)
Halogens (Cl_2, Br_2, I_2)	Room temperature	Dihaloalkane	$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$ (1,2-dibromoethane)

```
Addition of bromine to ethene (this is the bromine-water test):

    H     H                          H   H
     \   /                           |   |
      C=C      +    Br — Br    →    H — C — C — H
     /   \                           |   |
    H     H                          Br  Br

  ethene        bromine          1,2-dibromoethane
(unsaturated)  (orange)          (saturated, colourless)
```

Addition of hydrogen to propene (nickel catalyst, ~60 °C):

$C_3H_6 + H_2 \rightarrow C_3H_8$ propene → propane

Addition of steam to propene (phosphoric acid catalyst):

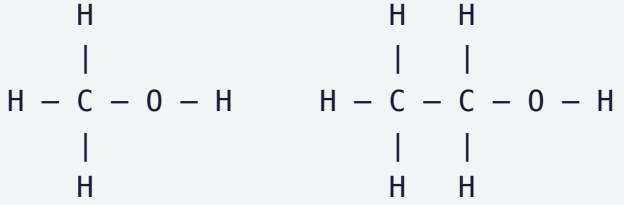
$C_3H_6 + H_2O \rightarrow C_3H_7OH$ propene → propanol

Exam Tip: Hydrogenation is used industrially to **harden vegetable oils** into margarine and spreads (this links to Biology). Know the catalyst: **nickel**. For hydration of ethene the catalyst is **phosphoric acid**. Mixing up the two catalysts is a common error in 1-mark recall questions.

4.7.2.3 Alcohols

Alcohols contain the functional group **–OH** (hydroxyl group). The general formula is $C_nH_{2n+1}OH$.

Name	Molecular formula	Condensed structural formula
Methanol	CH_3OH	CH_3OH
Ethanol	C_2H_5OH	CH_3CH_2OH
Propanol	C_3H_7OH	$CH_3CH_2CH_2OH$
Butanol	C_4H_9OH	$CH_3CH_2CH_2CH_2OH$



```
methanol          ethanol
```

Exam Tip: In a displayed formula, the –OH must be drawn as **C — O — H** with the bond from carbon going to the **oxygen**, not to the hydrogen. "C — H — O" is wrong and will not be credited.

Reactions of Alcohols

Reaction	What you observe / what forms	Equation (ethanol)
With sodium	Fizzing (hydrogen gas produced), sodium dissolves — but much more slowly than sodium in water. Forms sodium ethoxide.	$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$
Combustion in air	Burns with a clean, blue flame — used as a fuel.	$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$
With water	Methanol, ethanol and propanol dissolve to form a **neutral** solution (pH 7).	—
Oxidation	With an oxidising agent (e.g. acidified potassium dichromate, which turns orange → green) or by **microbial** action in air, ethanol is oxidised to **ethanoic acid**. This is why wine left open turns to vinegar.	$C_2H_5OH + 2[O] \rightarrow CH_3COOH + H_2O$

Uses of Alcohols

- **Fuels** — ethanol is blended with petrol (e.g. E10 petrol is up to 10% ethanol); methanol is used as a fuel in some racing cars.
- **Solvents** — ethanol dissolves many substances that water cannot (e.g. in perfumes, aftershaves, mouthwashes, inks); methanol is in some industrial solvents.
- **Alcoholic drinks** — the main alcohol in alcoholic drinks is **ethanol**.

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Making Ethanol by Fermentation

> Aqueous solutions of ethanol are produced when **sugar** solutions are **fermented using yeast**.
>
> `glucose → ethanol + carbon dioxide`
> $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

Conditions needed for fermentation:

- **Yeast** (contains the enzymes that catalyse the reaction).
- **Warm temperature** of about **30–35 °C** — too cold and the reaction is slow; too hot (above ~40 °C) and the yeast enzymes are **denatured**.
- **Anaerobic conditions** (no air) — if oxygen is present, the ethanol is oxidised to ethanoic acid (vinegar).
- **Aqueous** solution of sugar, at a suitable pH.

Fermentation stops when the ethanol concentration reaches about 14–15%, because the ethanol becomes toxic to the yeast. A more concentrated solution can be obtained by fractional distillation.

Comparing Fermentation with Hydration of Ethene

Feature	Fermentation	Hydration of ethene
Raw material	Sugar from crops (sugar cane, sugar beet) — **renewable**	Ethene from crude oil — **non-renewable**
Conditions	~30–35 °C, atmospheric pressure, yeast	~300 °C, ~60–70 atm, phosphoric acid catalyst
Process type	Batch — slow	Continuous — fast
Purity of product	Dilute (~14%) — needs distillation	Very pure ethanol

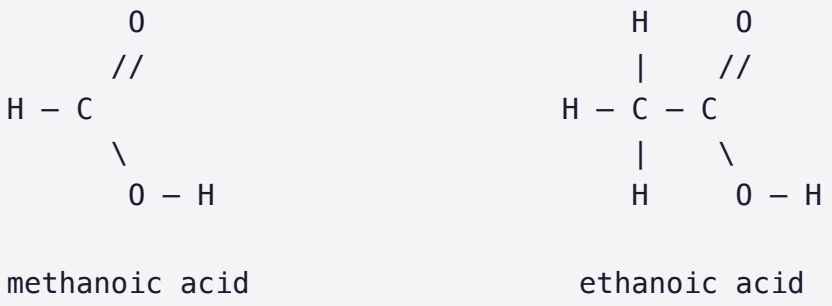
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Atom economy	~51% (CO_2 is a by-product)	100% (addition reaction, one product)

4.7.2.4 Carboxylic Acids

Carboxylic acids have the functional group **–COOH** (carboxyl group).

Name	Formula	Notes
Methanoic acid	HCOOH	Found in ant and nettle stings.
Ethanoic acid	CH_3COOH	The acid in vinegar.
Propanoic acid	CH_3CH_2COOH	Used as a food preservative.
Butanoic acid	$CH_3CH_2CH_2COOH$	Smell of rancid butter.



Exam Tip: The carbon in –COOH is counted in the name. Ethanoic acid (CH_3COOH) has **two** carbons in total, so it comes from ethanol (also two carbons). A common error is to name CH_3COOH "methanoic acid" because you see "CH_3".

Reactions of Carboxylic Acids

1. Dissolve in water to form **acidic** solutions (pH below 7), because they release H^+ ions.

2. React with carbonates like other acids, to give a salt, water and carbon dioxide (fizzing):

ethanoic acid + sodium carbonate → sodium ethanoate + water + carbon dioxide

$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$

The salts of ethanoic acid are called **ethanoates** (e.g. sodium ethanoate, CH_3COONa).

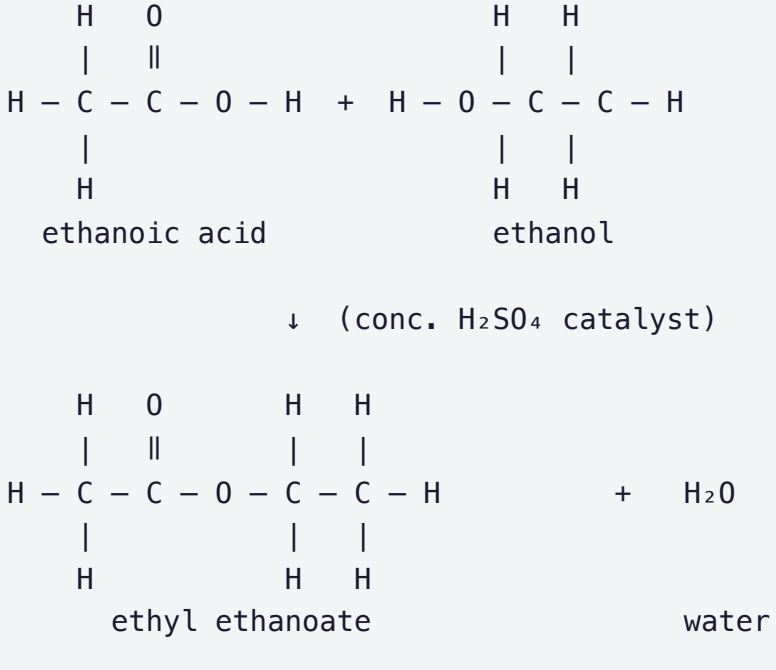
3. React with alcohols in the presence of an **acid catalyst** (a few drops of concentrated sulfuric acid) to produce **esters** and water:

carboxylic acid + alcohol → ester + water

ethanoic acid + ethanol → ethyl ethanoate + water

$CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$

Esters have the functional group **–COO–**. They are **volatile** compounds with distinctive, often **fruity smells**, and are used as **flavourings** and in **perfumes**. Ethyl ethanoate is also used as a **solvent** (e.g. in nail-varnish remover and glues).



Exam Tip: Naming esters: the part from the **alcohol** comes **first** and ends in **-yl**; the part from the **acid** comes **second** and ends in **-oate**. So ethanol + ethanoic acid → **ethyl**

ethanoate; methanol + propanoic acid → methyl propanoate. Only ethyl ethanoate is required by name, but the pattern is useful.

Carboxylic Acids Are Weak Acids (HT only)

Carboxylic acids are **weak acids**: they **only partially ionise** in aqueous solution. Most of the molecules stay as un-ionised CH_3COOH, and a reversible reaction sets up an equilibrium:

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

A strong acid such as hydrochloric acid **fully ionises**: HCl(aq) → H^+(aq) + Cl^-(aq).

So, for aqueous solutions of the **same concentration**, a carboxylic acid has a **lower concentration of H^+ ions** and therefore a **higher pH** than a strong acid. It also reacts more slowly with metals and carbonates (less vigorous fizzing).

Exam Tip: Do not confuse **weak** with **dilute**. "Weak" describes the degree of *ionisation*; "dilute" describes the *amount of acid per unit volume*. You can have a concentrated solution of a weak acid (e.g. glacial ethanoic acid) and a dilute solution of a strong acid.

Summary of Functional Groups

Homologous series	Functional group	General formula	Ending	Example
Alkanes	C–C only (none)	C_nH_{2n+2}	-ane	Ethane, C_2H_6
Alkenes	C=C	C_nH_{2n}	-ene	Ethene, C_2H_4
Alcohols	–OH	$C_nH_{2n+1}OH$	-anol	Ethanol, C_2H_5OH
Carboxylic acids	–COOH	$C_{n-1}H_{2n-1}COOH$	-anoic acid	Ethanoic acid, CH_3COOH

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Esters	–COO–	—	-yl -oate	Ethyl ethanoate, $CH_3COOC_2H_5$

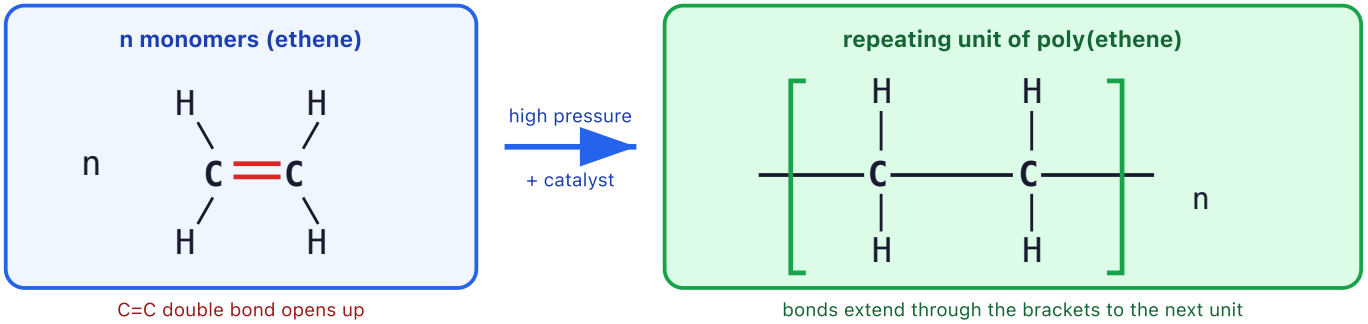
4.7.3 Synthetic and Naturally Occurring Polymers (Chemistry only)

4.7.3.1 Addition Polymerisation

> **Polymers** are very large molecules made by joining together many small molecules called **monomers**.
>
> In **addition polymerisation**, many **alkene** monomers join together. The C=C double bond in each monomer opens up and the monomers link into a long chain. The polymer is the **only product**.

In an addition polymer, the **repeating unit** has the **same atoms** as the monomer because no other molecule is formed in the reaction.

Addition polymerisation — ethene to poly(ethene)



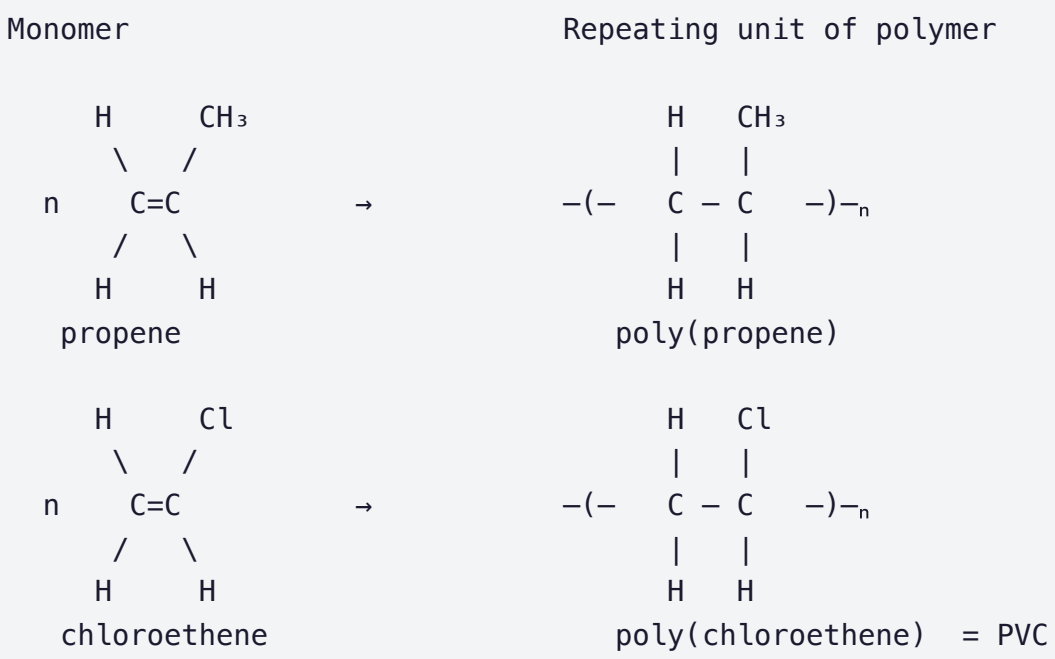
The repeating unit contains exactly the same atoms as the monomer — nothing is lost.

Drawing Repeating Units

You should be able to:

- draw the **repeating unit** from a given monomer, and

- work out the **monomer** from a given repeating unit (put the double bond back between the two main-chain carbons and remove the extension bonds and brackets).



Monomer	Polymer	Typical uses
Ethene	Poly(ethene) (polythene)	Carrier bags, bottles, cling film
Propene	Poly(propene)	Crates, ropes, carpets
Chloroethene (vinyl chloride)	Poly(chloroethene), PVC	Window frames, water pipes, cable insulation
Tetrafluoroethene	Poly(tetrafluoroethene), PTFE	Non-stick pan coatings

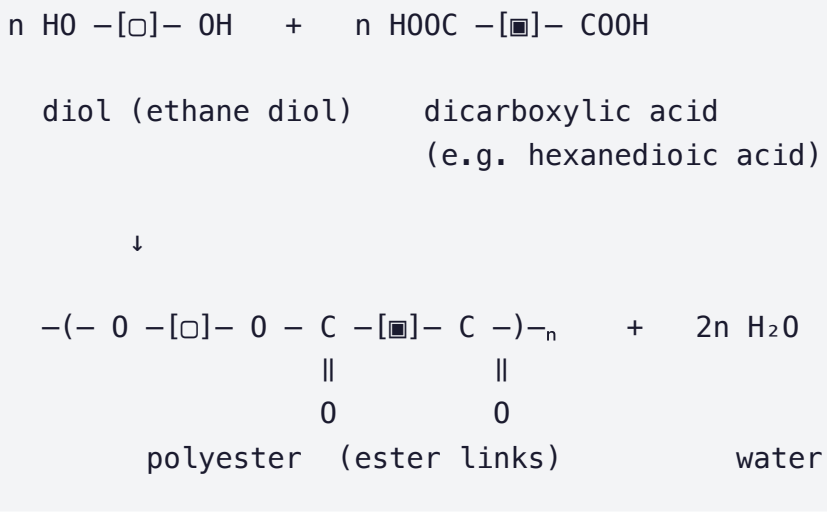
Exam Tip: When drawing an addition polymer repeating unit, three things earn the marks: (1) the C=C becomes a **single bond**; (2) **bonds extend outwards through the brackets** on both sides; (3) a subscript **n** after the closing bracket. Any side groups (CH_3, Cl) stay attached to the same carbons as in the monomer.

4.7.3.2 Condensation Polymerisation (HT only)

> **Condensation polymerisation** involves monomers with **two functional groups**. When these monomers react, they join together and **lose small molecules** such as **water** (or HCl). So there are **two products**: the polymer and the small molecule.

The simplest condensation polymers are made from **two different monomers**, each with two of the same functional group. For example, a **diol** (two –OH groups) reacts with a **dicarboxylic acid** (two –COOH groups) to form a **polyester**. Each link is an **ester** link (–COO–), and one molecule of water is lost each time a link forms.

Exam questions use a simplified representation in which the carbon chain between the functional groups is shown as a box:



Note that **2n** molecules of water are lost — each repeating unit contains two ester links.

Comparing Addition and Condensation Polymerisation

Feature	Addition polymerisation	Condensation polymerisation
Monomers	Alkenes — contain C=C	Two functional groups per monomer (e.g. –OH and –COOH)
Number of types of monomer	Usually one	Usually two different monomers (or one with two different groups, e.g. amino acids)

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Number of products	**One** — the polymer only	**Two** — the polymer and a small molecule (e.g. H_2O)
Repeating unit vs monomer	Same atoms as the monomer	Fewer atoms than the monomers combined (water removed)
Examples	Poly(ethene), poly(propene), PVC	Polyesters (PET, Terylene), nylon, proteins, starch

4.7.3.3 Amino Acids (HT only)

> **Amino acids** have **two different functional groups** in the same molecule: an **amine group ($-NH_2$)** and a **carboxylic acid group (–COOH)**.
>
> Amino acids react by **condensation polymerisation** to produce **polypeptides**. The $-NH_2$ of one amino acid reacts with the –COOH of the next, eliminating water.

The simplest amino acid is **glycine**, H_2NCH_2COOH:

```
glycine:          H   H   O
                   \  |  //
                    N–C–C
                   /  |  \
                  H   H   O–H

Polymerisation of glycine:

n H₂NCH₂COOH   →   –(– HNCH₂COO –)–ₙ   +   n H₂O
                       (shown as –(–NH–CH₂–C(=O)–)–ₙ)
```

Because one amino acid molecule has both functional groups, only **one type of monomer** is needed, and **n** water molecules are lost for n monomers. Different amino acids can be combined in the same chain to produce **proteins**. There are about 20 different naturally occurring amino acids; the **order** in which they are joined determines the protein's shape and function (e.g. enzymes, haemoglobin, keratin).

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> **Exam Tip:** Higher-tier questions often give you an unfamiliar monomer and ask you to say whether it forms an addition or a condensation polymer. Look for a **C=C** (addition) or for **two reactive groups** such as –OH, –COOH or $–NH_2$ (condensation, with loss of water).

4.7.3.4 DNA and Other Naturally Occurring Polymers

Many important biological molecules are polymers. You should be able to name the types of monomers from which these naturally occurring polymers are made.

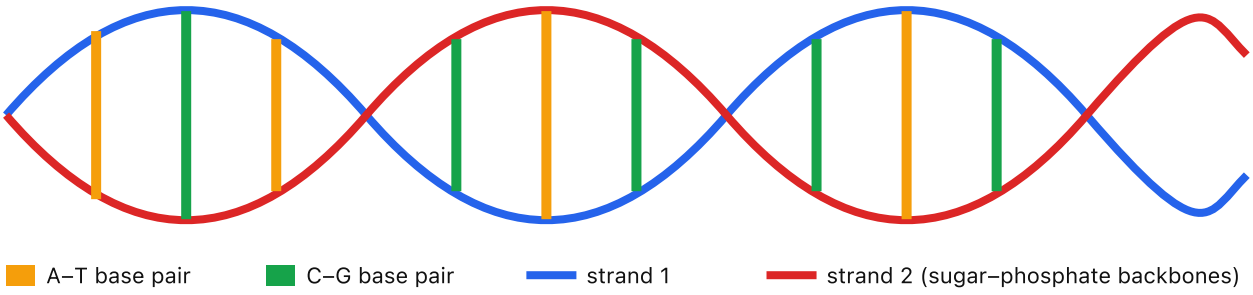
Natural polymer	Monomers	Notes
DNA (deoxyribonucleic acid)	**Nucleotides**	Encodes genetic instructions for the development and functioning of living organisms and viruses.
Proteins	**Amino acids**	Condensation polymers (polypeptides); enzymes, antibodies, muscle, hair.
Starch	**Glucose** (a sugar)	Energy store in plants.
Cellulose	**Glucose** (a sugar)	Structural material in plant cell walls.

The Structure of DNA

- DNA is a large molecule essential for life. It encodes the genetic instructions for the development and functioning of living organisms and viruses.
- Most DNA molecules are **two polymer chains**, made from **four different monomers** called **nucleotides**, in the form of a **double helix**.
- Each nucleotide contains a sugar, a phosphate group and one of four **bases**: adenine (A), thymine (T), cytosine (C) and guanine (G).
- The two strands are held together by weak bonds between complementary bases: **A pairs with T** and **C pairs with G**.

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DNA — two polymer strands of nucleotides (simplified)



Exam Tip: Exam questions on natural polymers are usually short recall: "Name the monomer of starch" (**glucose** / sugar), "Name the monomers in DNA" (**nucleotides**), "Name the monomers in proteins" (**amino acids**). Make sure you know which is which — and that DNA has **two** strands in a **double helix**.

Summary Checklist for 4.7 Organic Chemistry

- Describe crude oil as a finite mixture of hydrocarbons formed from ancient plankton.
- Recall the general formula of alkanes, C_nH_{2n+2}, and name and draw methane, ethane, propane and butane.
- Explain how fractional distillation separates crude oil, and name the main fractions and their uses.
- Describe how boiling point, viscosity and flammability change with chain length, using intermolecular forces.
- Write balanced equations for the complete combustion of hydrocarbons.
- Describe catalytic and steam cracking; balance cracking equations; explain why cracking is needed using supply/demand data.
- Describe the bromine-water test for alkenes (orange → colourless).
- *Chemistry only:* recall the general formula C_nH_{2n}, name and draw ethene, propene, butene and pentene.
- *Chemistry only:* describe the combustion and addition reactions of alkenes with hydrogen (Ni), steam (H_3PO_4) and halogens.

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- *Chemistry only:* name and draw methanol, ethanol, propanol and butanol; describe their reactions with sodium, air, water and oxidising agents; describe fermentation and its conditions.
- *Chemistry only:* name and draw the first four carboxylic acids; describe reactions with carbonates, water and alcohols (esters; ethyl ethanoate).
- *HT:* explain why carboxylic acids are weak acids (partial ionisation).
- *Chemistry only:* draw addition polymer repeating units from monomers and vice versa.
- *HT:* explain condensation polymerisation (diol + dicarboxylic acid → polyester + water) and how amino acids form polypeptides and proteins.
- *Chemistry only:* name the monomers of DNA (nucleotides), proteins (amino acids), starch and cellulose (glucose); describe the double-helix structure of DNA.

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